Lexis-AI: Neural Audio Restoration Engine

A professional technical project report documenting the architecture, team, and vision behind a deep-learning audio restoration system designed for digital education.

TECHNICAL REPORT

DEEPFILTERNET · FASTAPI · YT-DLP

Abstract

Lexis-AI is a specialised restoration engine designed to eliminate background noise from academic recordings, delivering pristine audio quality to learners worldwide. As digital education continues to expand across platforms and geographies, the demand for high-fidelity lecture content has never been greater — yet the audio quality of most online educational material remains inconsistent, with many recordings suffering from poor capture conditions that undermine their instructional value. Lexis-AI addresses this gap by applying state-of-the-art neural audio processing to transform noisy, degraded recordings into clear, intelligible speech.

At its core, Lexis-AI leverages **DeepFilterNet**, a state-of-the-art neural network architecture for real-time speech enhancement. The system integrates seamlessly with YouTube lecture sources via **yt-dlp** and exposes a clean, high-performance REST interface through **FastAPI**, enabling asynchronous processing at scale. DeepFilterNet was selected for its proven ability to generalise across diverse noise environments while maintaining low inference latency — a critical requirement for processing large volumes of educational content efficiently.

🎯 Purpose

Eliminate background noise from academic recordings to improve learner comprehension and engagement.

⚙️ Engine

DeepFilterNet delivers real-time, multi-frame spectral filtering for superior speech reconstruction.

🌍 Reach

Designed to serve global digital learners with crystal-clear, accessible audio content.

Problem Statement: The Education Noise Crisis

The rapid shift to digital education has exposed a critical infrastructure gap: audio quality. Lecture recordings captured in home offices, classrooms, and improvised studios frequently contain disruptive environmental noise that degrades the learning experience.

The Noise Problem

Digital lecture recordings often suffer from HVAC hums, room reverberation, keyboard clicks, and ambient white noise — all of which compete with the lecturer's voice for the student's attention.

Impact on Learning

- Low audio fidelity increases **cognitive load**, forcing students to work harder to decode speech
- Reduced information retention and engagement in prolonged listening sessions
- Automated transcription tools fail when **Signal-to-Noise Ratio (SNR)** drops below operational thresholds
- Students with hearing difficulties are disproportionately affected

~12dB

Typical SNR Loss

Average degradation in home-recorded lecture environments versus studio conditions.

40%

Retention Drop

Estimated reduction in information retention when audio quality is poor.

3×

Transcription Errors

Automated tools produce three times more errors in noisy audio conditions.

Technical Architecture: The Pipeline

Lexis-AI is built upon three tightly integrated components, each selected for performance, reliability, and compatibility within a modern Python ecosystem. The architecture follows a microservices-inspired pattern, enabling independent scaling of ingestion, processing, and delivery layers.

Backend Framework — FastAPI

Provides a high-performance REST interface for asynchronous request handling. FastAPI's native support for `async/await` patterns enables concurrent processing of multiple audio jobs without blocking the event loop, making it well-suited for production-grade audio workloads.

Source Retrieval — yt-dlp

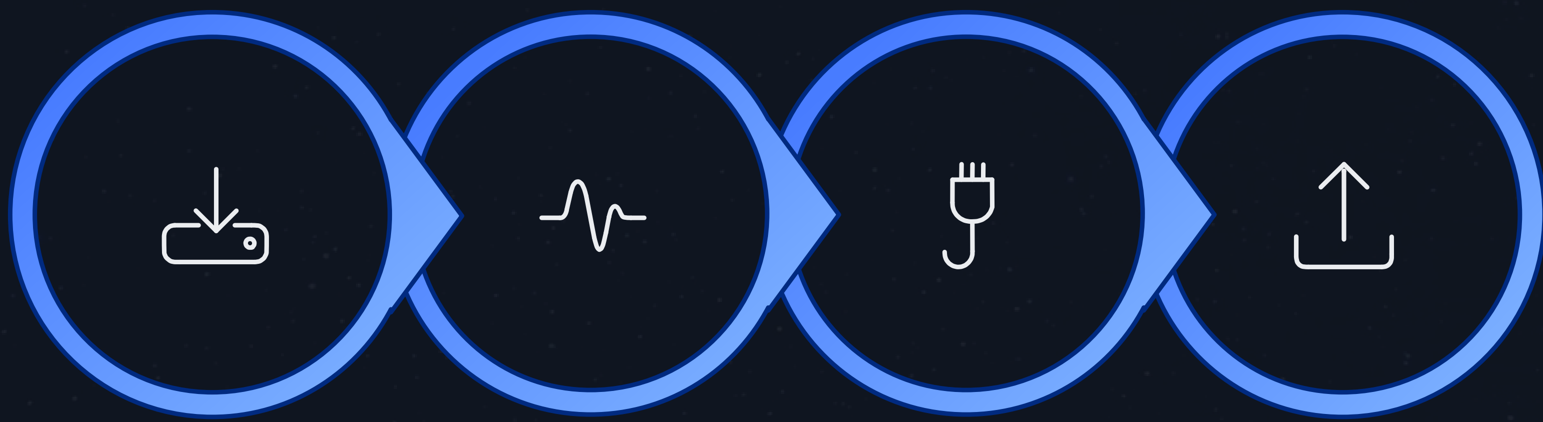
Automates secure ingestion of YouTube lecture content with full metadata extraction. The tool supports playlist parsing, format selection, and robust error handling for interrupted downloads, ensuring reliable acquisition of source material at scale.

Restoration Engine — DeepFilterNet

Executes multi-frame spectral filtering to isolate speech from chaotic environmental noise. DeepFilterNet operates in the frequency domain, applying learned filters across consecutive audio frames to reconstruct a clean speech envelope while preserving natural vocal characteristics.

Core Workflow

From raw YouTube URL to restored audio output, Lexis-AI follows a deterministic four-stage pipeline. Each stage is designed to be fault-tolerant, with comprehensive logging and error recovery mechanisms to ensure consistent output quality across diverse source material.



Ingestion | Preprocessing | Enhancement | Delivery

The pipeline is engineered for end-to-end latency of under 30 seconds for typical lecture segments, with each stage operating as an independent async task within the FastAPI event loop.

Stage 1 — Ingestion

yt-dlp downloads the source stream and converts it to a standardised 48 kHz WAV format, ensuring compatibility with downstream processing stages regardless of the original encoding.

Stage 2 — Preprocessing

Audio segments are normalised for **DeepFilterNet** input requirements, including amplitude scaling, silence trimming, and frame alignment to match the model's expected tensor dimensions.

Stage 3 — Enhancement

The neural engine applies complex spectral filters to reconstruct the clean speech envelope, suppressing non-speech components while preserving vocal timbre and prosody.

Stage 4 — Delivery

Processed audio is passed to downstream transcription pipelines or returned directly to the end-user via the REST API, with optional metadata including SNR improvement metrics.

Team Structure: The Execution Force

Lexis-AI is developed and maintained by a cross-functional team of six specialists, each responsible for a critical layer of the system. The team structure reflects the multidisciplinary nature of the project, spanning machine learning, backend engineering, data infrastructure, quality assurance, and user experience design.

Rudrakant Mishra — Project Lead

Manages timeline execution, GitHub coordination, final report documentation, presentation deck creation, and testing coordination across all pipeline stages.

Vinsh Kushwaha — Audio Extraction Engineer

Handles audio extraction from video sources, format conversion, raw data cleaning, and volume normalization to maintain consistent input quality.

Jeel Amrutiya — Signal Processing Specialist

Performs FFT analysis, spectrogram generation, frequency filtering, and detailed noise analysis across varied acoustic environments.

Harshit — Model Integration Engineer

Builds and deploys the DeepFilterNet denoising model, trains on diverse datasets, and evaluates restoration performance against quality benchmarks.

Manvi Singh — UI/UX Developer

Designs the interactive Streamlit interface, ensuring seamless user experience, accessibility compliance, and clear workflow presentation.

Lalit — API Setup & Integration Engineer

Architects the asynchronous FastAPI backend, manages system module integration, and optimizes REST endpoints for production deployment.

Engineering Challenges & Neural Solutions

Addressing the inherent complexities of audio processing requires innovative solutions that blend advanced machine learning with robust engineering practices. Lexis-AI tackles these challenges head-on to deliver superior audio restoration.

Problem: Harmonic Distortion

Traditional spectral gating creates 'musical noise' and robotic artifacts that degrade speech naturalness and listener experience.

Technical Solution

Integrated DeepFilterNet's Perceptual Loss function to maintain speech naturalness while suppressing background noise. This approach preserves vocal timbre and prosody characteristics.

Problem: Stream Ingestion

Handling variable bitrates from YouTube URLs without quality loss or format incompatibility issues across diverse source material.

Technical Solution

Implemented a robust yt-dlp + FFmpeg pipeline to force 16-bit PCM WAV extraction at 48 kHz, ensuring consistent input quality regardless of source encoding.

Problem: Computational Overhead

DNN inference can be prohibitively slow on standard CPUs, creating bottlenecks in batch processing and real-time applications.

Technical Solution

Optimized the FastAPI backend with asynchronous task queuing and model quantization techniques for faster processing, enabling concurrent job handling without latency degradation.

Future Scope

Lexis-AI's roadmap extends well beyond batch processing of recorded lectures. The team has identified three major capability expansions that will transform the engine from a post-processing tool into a comprehensive, real-time audio intelligence platform for education.

1

Phase 1 — Real-Time Streaming

Introduce low-latency streaming support for live webinars and virtual classrooms, enabling on-the-fly noise suppression during live lectures without perceptible delay.

2

Phase 2 — Speaker Diarisation

Integrate speaker diarisation models to identify and separate multiple lecturers within a single recording, enabling per-speaker audio enhancement and transcript attribution.

3

Phase 3 — Browser Extension

Deploy a lightweight browser-based extension for on-the-fly restoration during playback, bringing Lexis-AI's capabilities directly to learners without requiring server-side processing.

Research Direction: The team is also exploring integration with large language models for semantic audio correction — automatically fixing mispronounced technical terms and filling gaps in degraded speech segments.

Conclusion: Defining the Future of EdTech

> "By standardising audio clarity, we ensure that the quality of information dissemination matches the quality of the content itself."

Lexis-AI bridges the gap between raw, noisy inputs and high-quality educational experiences. In an era where digital lectures are the primary medium of knowledge transfer for millions of students, audio quality is not a luxury — it is a fundamental requirement for equitable access to education.

The Problem We Solved

Noisy, inaccessible lecture recordings that hinder learning and frustrate students across the globe.

The Technology We Built

A production-grade neural audio restoration pipeline powered by DeepFilterNet, FastAPI, and yt-dlp.

The Future We Envision

A world where every student, regardless of location or device, receives crystal-clear educational audio.

Lexis-AI represents a meaningful step forward in the standardisation of digital education infrastructure. By treating audio quality as a first-class engineering concern, the project sets a new benchmark for what learners should expect from online educational content — and demonstrates that modern deep learning techniques can deliver that standard at scale.

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